# Assumption. Evidence.

A concise, image-backed audit of 22 data and visualization assumptions. Two remain contested.

| 22 | 20 | 02 |
|---|---|---|
| assumptions reviewed | current treatments | contested assumptions |

## A1 Timestamp units

ASSUMPTION

Treat ts as Unix seconds even though the parquet type says timestamp[ms]; normalize each match to t = ts - first event.

SUPPORTING EVIDENCE

- Reading it as milliseconds places the dataset in **January 1970** and compresses it into **6 min 40 s**.
- Reading it as seconds yields **9-14 February 2026** and match spans of roughly **13-890 s**.

SOURCE p. 3

## A2 Event decoding

ASSUMPTION

Decode the binary event column as UTF-8 and stop the build if an unrecognized event occurs.

SUPPORTING EVIDENCE

- All observed event bytes decode; the audit identifies **eight event types**, including Position, Loot and KilledByStorm.

SOURCE p. 3

## A5 Evidence-led deduplication

ASSUMPTION

Remove exact file copies and proven doubled journeys; preserve other matching rows where repeated gameplay events are plausible.

SUPPORTING EVIDENCE

- **1 of 1,243 files** is an identical copy; three journeys repeat their full first half, so **190 rows** are removed.
- Other same-second repeats remain; if they are logging artifacts, loot rates would differ by **7.6-14.5%**.

SOURCE p. 8

# The minimap is not a guess

Projection is validated mathematically and checked against visible map features.

## A3 Coordinate projection

ASSUMPTION

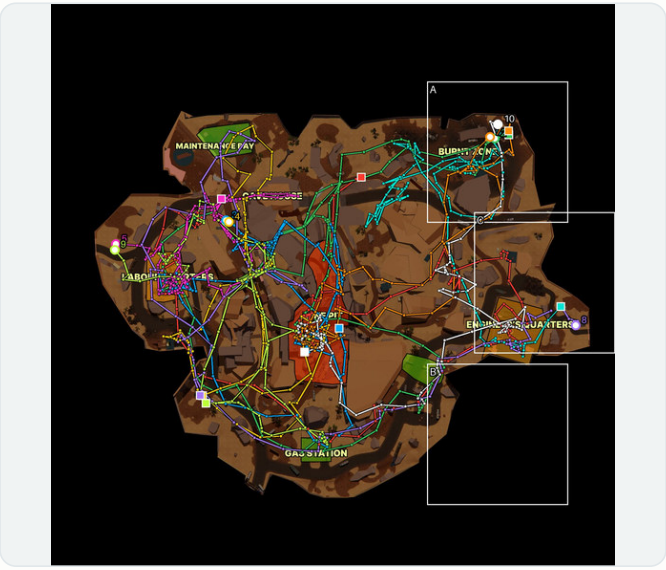
Convert world (x, z) to resolution-independent UV; draw y as (1 - v) x image height. Ignore world y (elevation).

SUPPORTING EVIDENCE

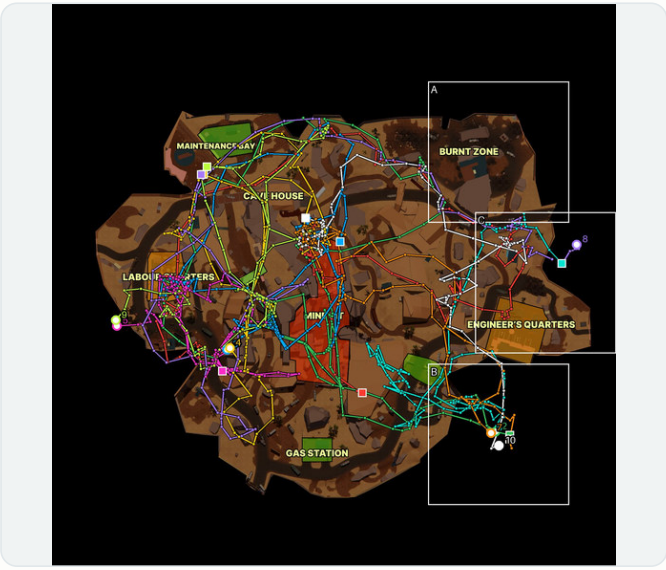
- The README example gives approximately **(78, 890)** at 1,024 px; the real source maps are **4,320² / 2,160 x 2,158 / 9,000²**.
- Independent audit: **100.00% / 99.69% / 99.99%** of samples land on map content for AmbroseValley / GrandRift / Lockdown.

**Limit:** No landmark ground truth; mapping is supported to about +/-10 world units, not proven pixel-perfect.

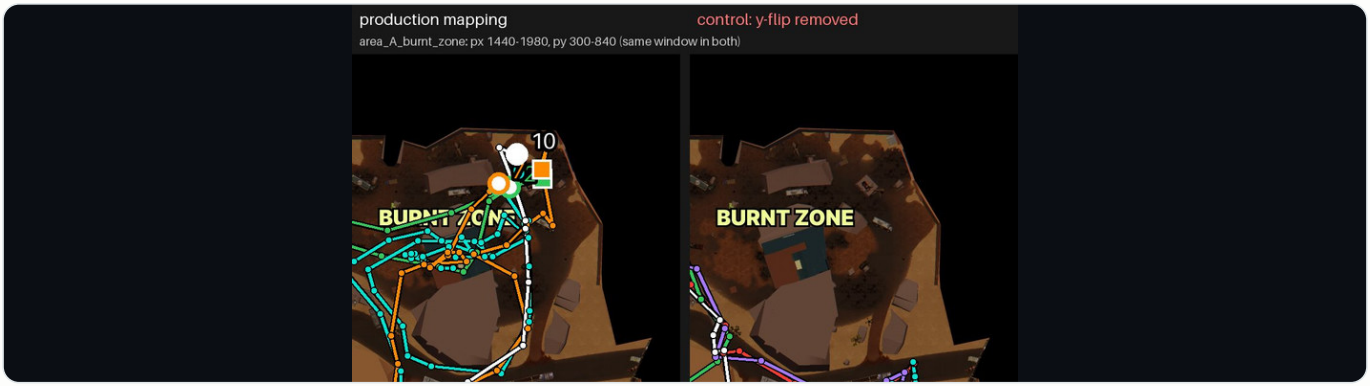
SOURCE pp. 3-7




Production: y-axis flipped correctly




Control: no y-flip




Original evidence: at Burnt Zone, starts align on map ground only with the production mapping (p. 5).

# Reversibility, checked

The browser and preprocessing code must agree on one coordinate system.

## A4 Reverse mapping

ASSUMPTION

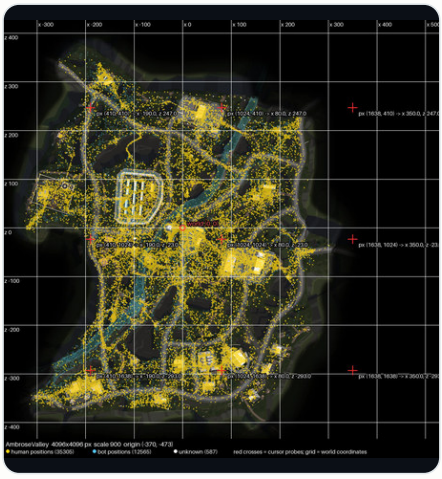
Use the inverse of the UV transform for map hover and location checks, with separate image width and height.

SUPPORTING EVIDENCE

- All **72,676 exported path points** round-trip to raw world coordinates within **0.0707 units** (the JSON rounding limit).
- **2,005 pixel probes per map** differ only by floating-point noise; GrandRift uses its non-square aspect ratio.

**Limit:** The source audit did not directly exercise deck.gl BitmapLayer placement in a running browser.

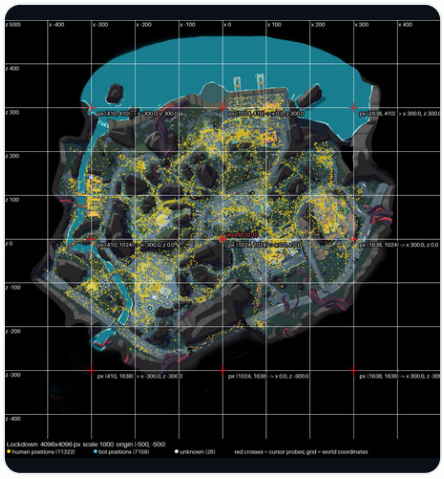
SOURCE pp. 7-8



AmbroseValley - coordinate grid




GrandRift - coordinate grid



Lockdown - coordinate grid

### Why this matters

One projection function ensures that journey paths, event markers, playback positions and heatmap locations cannot quietly disagree after resizing or zooming.

# What each record means

These rules prevent plausible-looking but misleading player journeys.

## A6 Human vs bot classification

ASSUMPTION

Classify an actor as a bot only when its file logs BotPosition; flag numeric-ID exceptions as id_mismatch.

SUPPORTING EVIDENCE

- **17 journeys** have numeric IDs but player-client events; **14** include Loot and **12** exceed the observed bot speed cap.
- Counter-evidence exists: one actor is logged as a bot in other matches. Treating all 17 as human changes any heatmap total by at most **1.63%**.

SOURCE pp. 8-9

## A7 Event position belongs to the actor

ASSUMPTION

A kill is positioned where the killer stood; a death where the recorded victim stood - not where an unobserved opponent stood.

SUPPORTING EVIDENCE

- Cross-file human BotKill and bot BotKilled pairings are a median **12.6 world units apart**, supporting owner-specific coordinates.

SOURCE p. 9

## A8 Unknown Kill/Killed cause

ASSUMPTION

Keep all three same-second Kill/Killed pairs, label the cause unknown, and exclude the paired Kill from kill totals.

SUPPORTING EVIDENCE

- Each pair shares time and coordinates but has **no recorded opposing actor**; the dataset cannot distinguish self-kill from a trade.
- Since **743 of 796** matches contain only one file, absence of a second actor is not proof of absence.

SOURCE p. 9

# Events and dates

Different forms of uncertainty need different handling rules.

## A9 No movement after death

ASSUMPTION

Drop Position samples occurring after the first death; retain late combat events with a flag.

SUPPORTING EVIDENCE

- **173 positions across 172 journeys** are excluded; **two later BotKill events** remain flagged after_death.

SOURCE p. 10

## A10 Date from the first event

ASSUMPTION

Use the UTC date of a match's first observed event, not the name of the folder holding the file.

SUPPORTING EVIDENCE

- **11 files** disagree with their folder date; **four matches** change date under this rule. First and last observation days are marked partial.

SOURCE p. 10

## A12 Bot recording is incomplete

ASSUMPTION

Say that bot journeys were recorded in 6.5% of matches; do not infer bots were absent from the others.

SUPPORTING EVIDENCE

- Only **52/796 matches** have a bot file, while **722/796** have human-file events recording bot combat.

SOURCE p. 11

# The traffic model is contested

The audit identified a material limitation, and the proposed fix is not applied.

## A11 Sampling gaps and traffic

CONTESTED

ASSUMPTION

Current behavior: split paths at gaps over 30 s; estimate traffic along shorter straight-line steps; omit longer gaps.

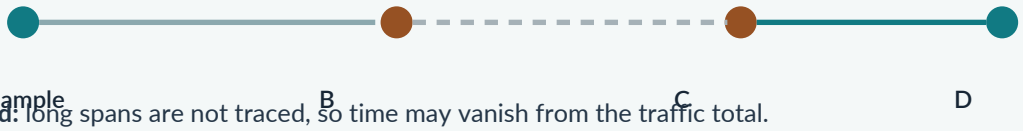
SUPPORTING EVIDENCE

- **99.92%** of long human sampling gaps align with the 5-second recording tick, and event positions cluster near the earlier point.
- The audit infers that missing samples often mean the actor was stationary: the production rule omits **3.9-4.3%** of human time.
- Overall traffic maps remain similar (**Spearman 0.983-0.996**) and the top traffic region on each map is unchanged across six models.

**Limit:** Suggested alternative: allocate (gap - 5 s) near the previous point. The source states this has NOT been implemented.

SOURCE pp. 10-11

### What the two models imply




**Implemented:** long spans are not traced, so time may vanish from the traffic total.

**Proposed:** keep potential stationary time at the last position (not implemented).

Conceptual sketch only; not an observed player trajectory.

# What the numbers can say

Labels distinguish directly observed events from interpretations.

## A13 Rates use recorded journeys

ASSUMPTION

Report rates per recorded human journey, not per match, when describing human outcomes.

SUPPORTING EVIDENCE

- **743 of 796** matches have one recorded file. A cross-page mismatch was corrected to **"kills per human journey"**.

SOURCE p. 11

## A14 Storm deaths are still deaths

ASSUMPTION

Include storm deaths within the Deaths heatmap, with storm counts displayed separately.

SUPPORTING EVIDENCE

- **39 storm deaths** are recorded; all occur **655-887 s** into a match, much later than the peak for bot-caused deaths.

SOURCE pp. 11-12

## A15 Ending alive is not extraction

ASSUMPTION

Describe the end of an unfinished journey as its last recorded point - never a confirmed extraction.

SUPPORTING EVIDENCE

- **339 journeys** end without a recorded death; the shortest lasts **86 s**. There is no extraction event in the source schema.

SOURCE p. 12

# Why hotspots use regions

The hotspot layer uses a larger spatial unit and a rule that is easy to audit.

## A16 64 x 64 cells are too sparse

ASSUMPTION

Retain 64 x 64 values for heatmap rendering but identify hotspots on 16 x 16 regions.

SUPPORTING EVIDENCE

- In a split-half test of AmbroseValley kills, top-5 overlap rises from **0.0 at 64 x 64** to **0.8 at 16 x 16**.

SOURCE p. 13

## A17 Simple match-based hotspot rule

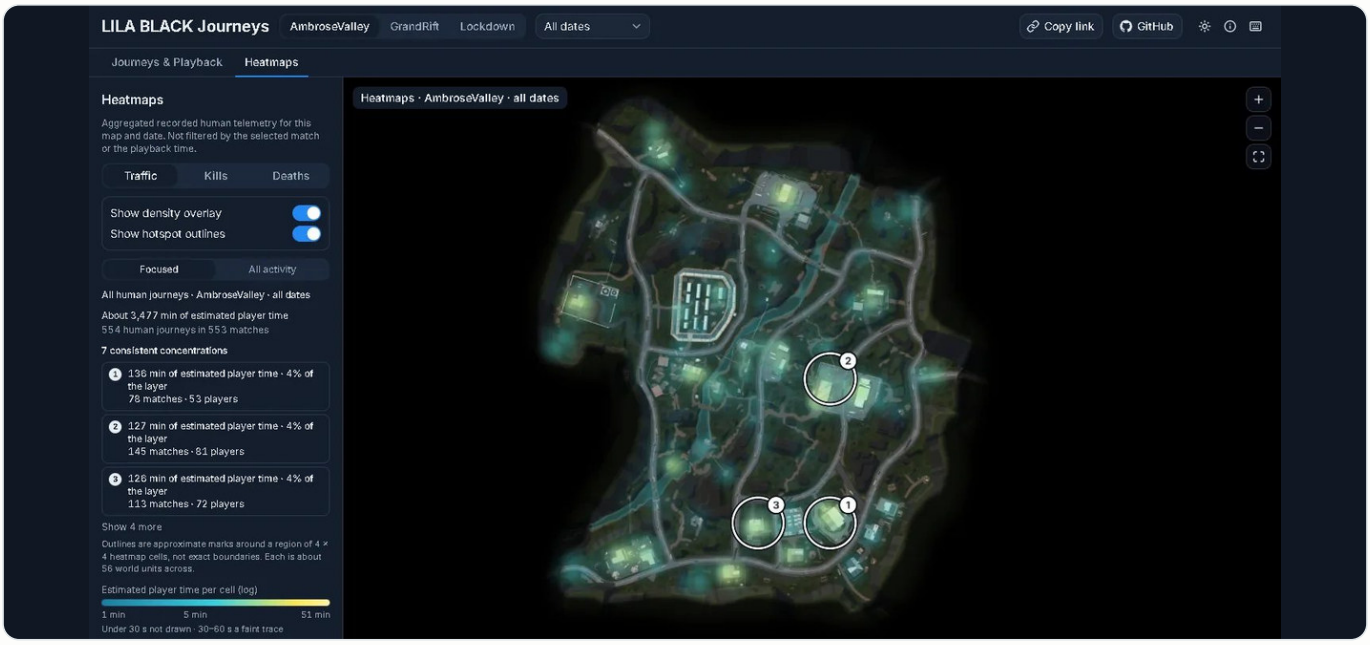
ASSUMPTION

Show up to six strongest regions when at least five matches contribute and no single match provides over half the value.

SUPPORTING EVIDENCE

- The earlier resampling rule changed which regions qualified with its random seed; its AmbroseValley death hotspot was selected in only **39% of 100 seed runs**.
- The final rule is explainable, but does **not** prove statistical stability or eliminate sensitivity to grid placement.

SOURCE pp. 13-15




Actual hotspot-outline UI. Numbered circles highlight qualifying regions; visual evidence from the source PDF (p. 14).

# Color should communicate counts

Display decisions are documented with both measurements and before/after images.

## A18 Use fixed, readable count classes

ASSUMPTION

For kill and death cells, use common bins of 1 / 2-3 / 4-9 / 10+ across maps and dates instead of a map-relative continuous gradient.

SUPPORTING EVIDENCE

- Under simulated color-vision deficiency, adjacent classes remain at least **18.9 delta-E** apart for kills and **7.0** for deaths.
- Continuous ramps could make differing counts effectively indistinguishable (GrandRift death colors reached **0 delta-E**).

SOURCE pp. 15-17



Kill heatmap - continuous -> hue-only bands -> final lightness-ordered count classes.



Death heatmap - the same three alternatives, showing why distinct count classes were necessary.

# Readable without changing data

The visual smoothing changes the display, not the counts behind the hover.

## A19 Render soft heatmap bumps

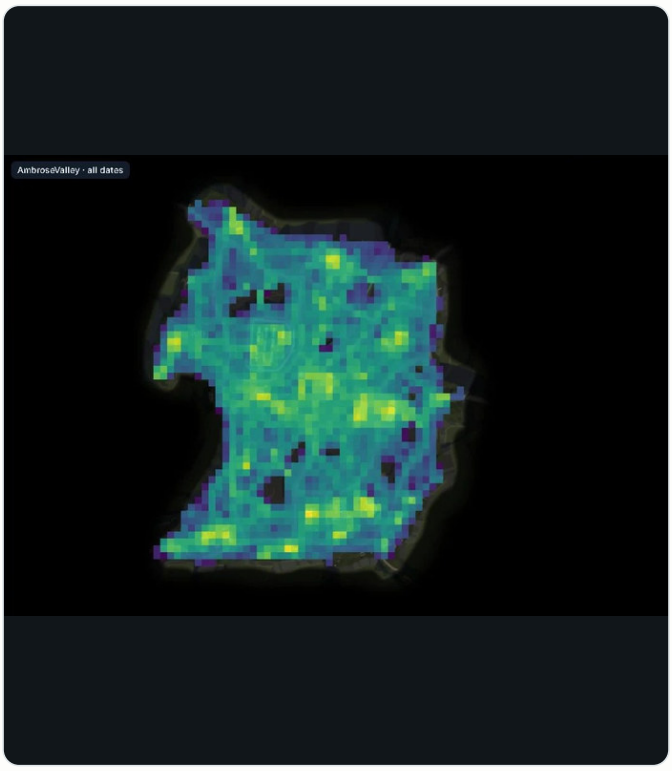
ASSUMPTION

Paint each cell as a soft bump but keep the strongest overlapping value, never sum neighboring values into a false peak.

SUPPORTING EVIDENCE

- The original mosaic obscured the minimap; the redesigned surface makes routes and hotspots easier to read.
- The underlying **64 x 64 grid and hover values remain unchanged** by the display-only rendering.

SOURCE pp. 18-19




Before: opaque grid mosaic




After: soft display-only rendering

Both examples are crops of the original screenshots (p. 19), with the application sidebar removed for readability.

# What holds, and what remains open

The implementation was extensively cross-checked; interpretation still needs judgment.

## A20 Composite map score is not quality

CONTESTED

ASSUMPTION

A five-part, equally weighted relative score can rank these three maps, but must not be presented as an objective quality measure.

SUPPORTING EVIDENCE

- The ranking is unchanged in **82.2%** of bootstraps with the published components; remove journey volume and it holds in only **56.5%**.
- The audit found broad uncertainty for GrandRift and dependence on which components are included.

**Limit:** Treat as a labeled relative index or remove "Leading" / "Needs improvement" language.

SOURCE p. 20

## A21 Rendered figures match pipeline outputs

ASSUMPTION

The displayed figures are expected to agree exactly with their generated JSON sources.

SUPPORTING EVIDENCE

- **8,011 internal data checks** and **815 rendered-page checks** found zero mismatches after a denominator-label defect was fixed.

SOURCE p. 20

## A22 Independent reproducibility

ASSUMPTION

A clean rebuild should give identical data artifacts, while separate audit logic independently checks numerical outputs.

SUPPORTING EVIDENCE

- The independent review reproduced all **796 match-index rows**, every match file, kill/death grid cells and **7,792/7,792** statistics.
- The uploaded audit document records **125 passing tests** at its audit point; subsequent project documentation reports newer test runs.

SOURCE p. 20